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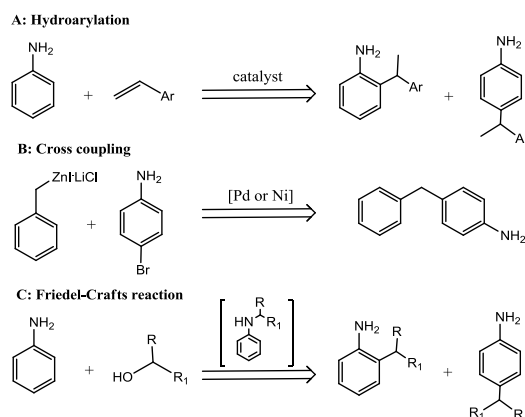
Palladium-Catalyzed Regioselective C-Benzylolation via a Rearrangement Reaction: Access to Benzyl-Substituted Anilines

Manuel Amézquita-Valencia, and Howard Alper*

This paper is dedicated to Prof. Dr. Helmut Schwarz, Berlin, Germany, in celebration of his wonderful contributions to science, policy, administration (A. von Humboldt Foundation), and to society. Integrity is his signature.

Abstract: An unprecedented C-benzylolation rearrangement reaction, catalyzed by palladium, is reported in this contribution. The reaction proceeds by rearrangement leading to the direct synthesis of *para* or *ortho* benzyl-substituted N-methylanilines. The reaction product is obtained in high regioselectivity, without the need to use a ligand for the catalytic process.

Benzyl-substituted anilines are important compounds employed in the design of organic compounds and pharmaceutical products.^[1] Consequently, the development of efficient and selective methods for the construction of amine substituted diarylmethanes, is a worthwhile goal in organic synthesis. An approach for the preparation of these type of compounds involves C-benzylolation via hydroarylation of anilines with styrene (Scheme 1A).^[2] There are very few examples of the use of N-methyl anilines as substrates, for the synthesis of the desired products.^[3] Other strategies used include the reduction of diaryl ketones containing the amino group in the structure,^[4] or cross coupling reactions catalyzed by transition metals (Scheme 1B).^[5]



Scheme 1. C/N-Benzylolation of Anilines

One of the methods used to obtain diarylmethanes is via the traditional Friedel-Crafts (FC) reaction, in which alkyl or aryl halides and a Lewis acid are utilized.^[6] However, amines are incompatible here, due to nitrogen coordination to the Lewis acid catalyst. In addition, mixtures of regioisomeric products are often

obtained in these reactions.^[2a,6b-c] In particular, the ability to form benzyl-substituted anilines from N-alkyl or aryl anilines is generally known as the Hofmann-Martius rearrangement (Scheme 1C, Friedel-Crafts reaction via H-M rearrangement).^[7] In general, the reaction is a thermal conversion of N-alkylanilines, involving the migration of an alkyl group in different ways: dependent on the nature of the olefin, radical or carbocation.^[7b-d] Likewise, the H-M rearrangement can be triggered by photo-irradiation.^[8]

In recent times, another way to address this issue is to use benzyl alcohol as the reagent instead of alkyl or aryl halides used traditionally in the FC reaction (Scheme 1C).^[9] Despite these achievements, we believe there are significant opportunities to develop new synthetic methodologies for the preparation of benzyl-substituted anilines. In addition, a practical catalytic procedure that is both chemo- and regioselective is meritorious. Herein, we wish to report a fascinating palladium-catalyzed rearrangement under CO pressure in methylene chloride, affording benzyl-substituted anilines in good yield and excellent selectivity. This transformation is carried out under catalytic conditions and the selectivity is sensitive to the amino substitution pattern. To our knowledge, this type of reactivity has never been reported before, and there only has been a report where n-substituted carbazoles in presence of phenol and H₃PO₄ resulted in the corresponding migration of a benzyl group.^[10]

Initially, 2-[(2-bromophenyl)methylamino]methyl]phenol **1a** was chosen as the model substrate for the rearrangement reaction. The precursors for our study were easily obtained based on the method of Mestichelli et al. via a four step sequence in high overall yield, and only the final product was purified by chromatography.^[11] The optimization study was initiated with 1 equivalent of K₃PO₄, and 2 mol % of palladium precursor under CO pressure, with dichloromethane as the solvent (Table 1). Initially, low product yield resulted when the reaction was carried out with a monodentate phosphine ligand, along with small amounts of byproducts (Entry 1-2). The reaction was totally ineffective using a bidentate ligand [e.g., 1,4-bis(diphenylphosphino)butane (dppb)], and the starting material was recovered (Entry 3). We were gratified to find that the Pd-catalyzed reaction occurred in the absence of a ligand, affording the rearrangement product **1b** in good yield and excellent selectivity (entry 4). The structure of **1b** was confirmed by single-crystal X-ray diffraction.^[12] Use of other solvents such as THF, MeCN and toluene gave inferior results, and no reaction occurred using coordinating solvents (Entries 5-7). Further testing of the palladium catalyst revealed that Pd(OAc)₂ is the best choice. The use of Pd(dba)₂ or PdCl₂ gave the product **1b** in low yield, while the yield was moderate using Pd(tfa)₂ (Entries 8-10).

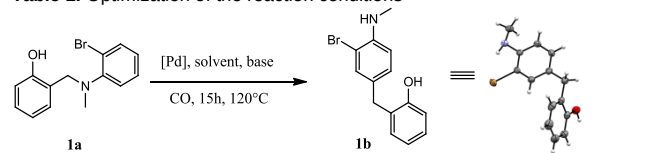
Although the reaction yield is good using K₃PO₄, different bases were examined to attempt to increase the yield of the reaction. Organic bases facilitate the decomposition of the substrate (Entries 12-15). Decreasing CO pressure proved detrimental to the yield of the product. In contrast, the yield was slightly enhanced when the pressure was increased to 600 psi

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(entries 16-18). Lowering the reaction temperature from 120 to 100 degrees gave recovered starting material (entry 19). Control experiments show that, in the absence of the palladium catalyst, no reaction occurs; a similar result is obtained in the absence of the base (entries 20-21).

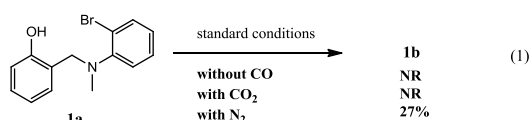
Table 1. Optimization of the reaction conditions^[a]



Entry	Catalyst	Base	Solvent	Yield % ^[b]
1 ^[c]	Pd(OAc) ₂	K ₃ PO ₄	DCM	40
2 ^[d]	Pd(OAc) ₂	K ₃ PO ₄	DCM	32
3 ^[e]	Pd(OAc) ₂	K ₃ PO ₄	DCM	s.m.
4	Pd(OAc) ₂	K ₃ PO ₄	DCM	83
5	Pd(OAc) ₂	K ₃ PO ₄	THF	s.m.
6	Pd(OAc) ₂	K ₃ PO ₄	MeCN	s.m.
7	Pd(OAc) ₂	K ₃ PO ₄	toluene	56
8	Pd(dba) ₂	K ₃ PO ₄	DCM	36
9	PdCl ₂	K ₃ PO ₄	DCM	23
10	Pd(tfa) ₂	K ₃ PO ₄	DCM	47
12	Pd(OAc) ₂	Na ₂ CO ₃	DCM	62
13	Pd(OAc) ₂	CSF	DCM	s.m.
14	Pd(OAc) ₂	Et ₃ N	DCM	-
15	Pd(OAc) ₂	DBU	DCM	-
16 ^[f]	Pd(OAc) ₂	K ₃ PO ₄	DCM	67
17 ^[g]	Pd(OAc) ₂	K ₃ PO ₄	DCM	65
18 ^[h]	Pd(OAc) ₂	K ₃ PO ₄	DCM	85
19 ^[i]	Pd(OAc) ₂	K ₃ PO ₄	DCM	s.m.
20	-	K ₃ PO ₄	DCM	s.m.
21	Pd(OAc) ₂	-	DCM	13

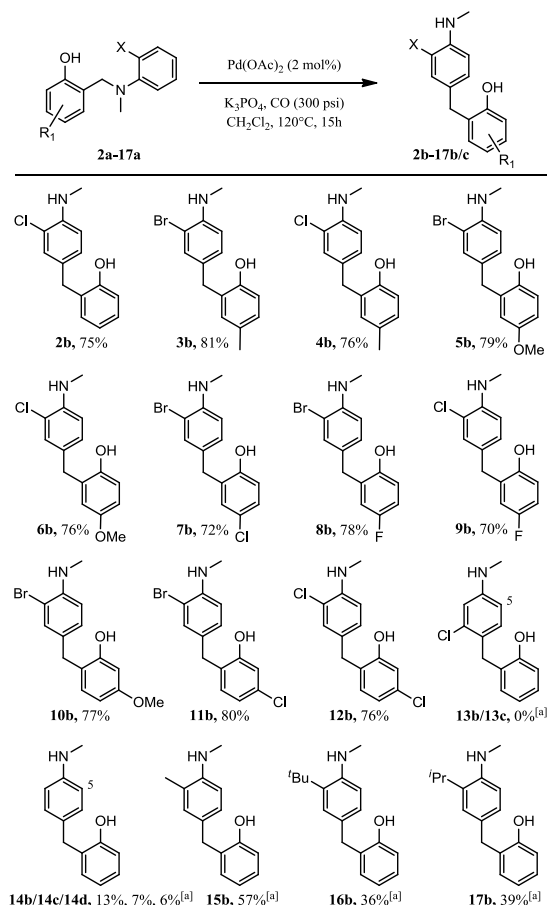
[a] Reaction conditions: **1a** (0.35 mmol), base 1 equivalent, 2 mol% [Pd] (0.0068 mmol), 300 psi of CO, 5 mL of solvent at 120°C. [b] After column chromatography. [c] PPh₃ was used (4 mol%). [d] Cytop 292 was used (4 mol%). [e] dppb was used (1 mol%). [f] 200 psi of CO. [g] 100 psi of CO. [h] 600 psi of CO. [i] Reaction at 100°C.

A set of experiments were carried out to explore the participation of gases in the reaction. First, the desired product was not formed in the absence of CO. Also, using carbon dioxide instead of CO gave recovered starting material while reaction under N₂, afforded **1b** in 27% yield (eq 1).



With optimum conditions in hand, we next explored the scope of the palladium-catalyzed rearrangement. In all cases, the reaction worked well using -Br or -Cl as a substituent on the amino containing moiety (Scheme 2, **2b** vs. **1b** Table 1, entry 4). For the N-biaryl compounds, bearing an electron-donating group such as methyl or methoxy group at the *para*-position with respect to the hydroxyl group, afforded the products in good yield (**3b-6b**). Similar results were observed with electron-

withdrawing groups in the same position (**7b-9b**). Changing the position of the substituent on the phenol moiety did not affect the yield (**10b-12b**). In contrast, changing the position of halogen did not give any product (**13b/13c**). On the other hand, absence of a substituent on the amino moiety produced a mixture of *para*- and *ortho*- isomers in low yields, 13% and 7% respectively (**14b/14c**), and the product of double benzylation was observed (**14d**). Changing the nature of the *ortho* substituent on the amino moiety gave the expected products in excellent selectivity, and in moderate yields (**15b**, **16b** and **17b**).

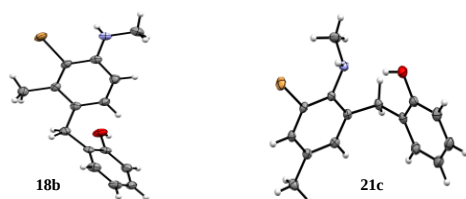
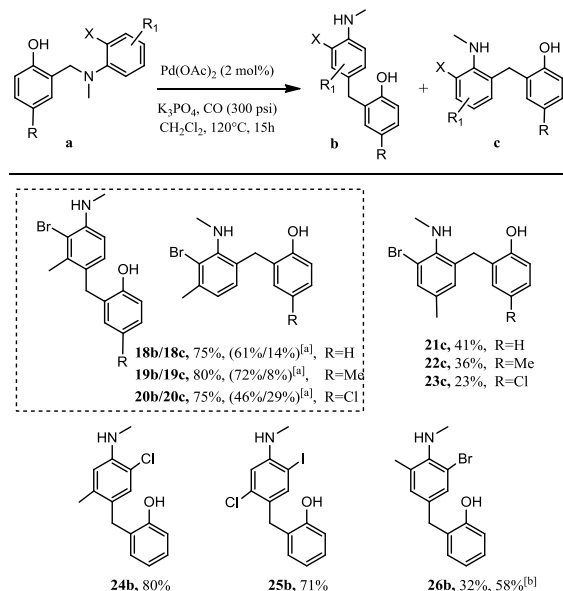


Scheme 2. Substrate Scope of N-Biaryl Derivatives. Reaction conditions: **2a-17a** (0.35 mmol), 2 mol % of Pd(OAc)₂ (0.0068 mmol), 1 equivalent of K₃PO₄, CO (300 psi), DCM (5 mL), 120°C, 15h. [a] The reaction was carried out at 600 psi of CO.

To explore the effect of the substituent on the amino containing moiety, a variety of substrates were synthesized, where the position of the methyl group around the aromatic ring was changed. These substrates afforded the expected products in moderate to good yields. The selectivity of the reaction was sensitive to the substitution pattern (Scheme 3).

We found that the methyl group next to halide directly influences the selectivity, affording a mixture of isomers in good yield, but in moderate selectivity (**18b/18c**). Furthermore, the substrate with an electron-donating methyl group on the alcoholic arm gave the products in both good yield and selectivity, 72%/8% (**19b/19c**), while an electron-withdrawing group (e.g., Cl) afforded low product selectivity, 46%/29% (**20b/20c**). In general, the selectivity of the reaction may be due to the presence of two directing groups (-NRR' and -Me) in the same aromatic ring, generating competition in the benzylation step. With the methyl group in the *para* position, only one

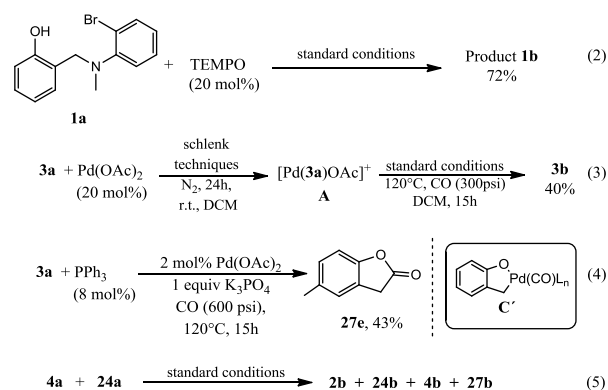
product was observed (**21c-23c**). Furthermore, the reaction with *meta*-substituted anilines resulted in the formation of only one regioisomer in good yield (**24b-25b**). Finally, the same result was found with the methyl group in the *ortho*-position (**26b**). The structures of the regioisomers **18b** and **21c** were confirmed by single-crystal X-ray diffraction (Figure 1).



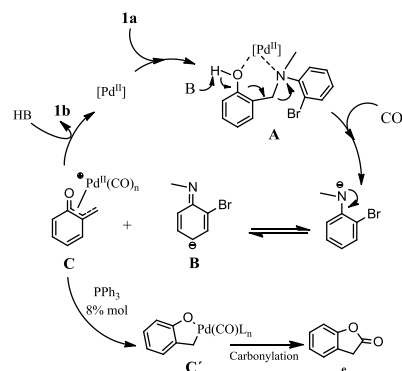
Several experiments were undertaken to gain some mechanistic insight into the present transformation. First, under standard reaction conditions using **1a**, four mercury drops were added in order to check the homogeneity of the reaction. The result showed no inhibition of the process and no significant differences in yield and selectivity were observed. This result is consistent with a homogeneous process. In addition, the presence of 2,2,6,6-tetramethylpiperidine 1-oxyl (TEMPO) did not inhibit the reaction, thus suggesting that a radical process is not involved (eq 2). Notably, when a stoichiometric reaction between $\text{Pd}(\text{OAc})_2$ and **3a** was carried out under nitrogen at room temperature, $[\text{Pd}(\text{3a})\text{OAc}]^+$ complex was found (eq 3). The formulation of the new complex was based on the ^1H -NMR and ESI experiments. Several unsuccessful attempts were made to fully characterize the complex, including growing suitable crystals for X-Ray analysis. When the reaction was conducted using complex **A**, the rearrangement product **3b** was formed in 40% yield. The latter result may indicate a chelation-assisted pathway. This behavior is supported by the results obtained in Table 1 (entries 1-3 vs. 4), when a negligible amount of product was formed by the action of the ligand in the reaction. The phosphine was detrimental to the reaction; retarding the

substrate coordination to the metal center. Furthermore, changing the position of the substituents (OH and Br) on the substrate failed to generate the rearrangement reaction; identical results were obtained when the substituents were removed. These experiments suggest that the position of the hydroxyl group is very important for the formation of the initial complex **A**.

An additional reaction was carried out using 8 mol% of PPh_3 under standard conditions at 600 psi of CO. This reaction was modestly effective for the rearrangement process, giving the carbonylation product **27e** (eq 4). This indicates that the formation of the new compound is by CO insertion, which means that the species **C** and **C'** might be involved in the process. Similar species have been reported for the functionalization of vinyl phenols and alkenes.^[13] In order to check if the process is inter or intramolecular, the reaction was carried out using a mixture of two different substrates **4a** and **24a** under optimized reaction conditions. As shown in equation 5, four compounds were detected by GC/MS (see Supporting Information, Figure S1). It is conceivable that an intermolecular process is operative here.



Although the precise mechanism for this migration reaction is yet to be fully elucidated, on the basis of the above results, a possible mechanism is outlined in Scheme 4. Initially, the reaction between $\text{Pd}(\text{OAc})_2$ and substrate **1a** may be chelation-directed, resulting in the formation of complex **A**.^[14] Subsequently, the base removes the proton from the hydroxyl group, releasing the ionic species **B**.^[15] Simultaneously, intermediates **C** and **C'** may be formed, in which case they could play a pivotal role in the reaction.^[13,16] At this point, CO coordination can help to stabilize different palladium species present in the reaction. Under conditions shown in equation 4, the formation of the palladacycle **C'** followed by CO insertion may afford the product **27e** by cyclocarbonylation. The formation of the carbonylation product is associated to excess ligand; increasing electronic density in the metal center decreases the possibility of a nucleophilic attack. Under standard conditions (ligand free), the C-C bond formation might be derived from Michael addition of **C** by the intermediate **B**, in which both *para*- and *ortho*-carbon of **B** could act as nucleophile, affording the rearrangement product **1b**. In addition, the generation of *ortho*-product might be obtained via stabilizing coordination of nitrogen.



Scheme 4. Proposed Mechanism

In conclusion, we have developed a novel and efficient Pd-catalyzed rearrangement of N-biaryl substrates for the direct synthesis of benzyl-substituted N-methylamines under CO and ligand-free conditions. In contrast, the presence of a ligand leads to the formation of the carbonylation product. The selectivity of the process can be controlled by the amino substitution pattern. This reaction has significant synthetic utility, particularly as a consequence of the high regioselectivity of the process.

Acknowledgements

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Keywords: Palladium • Rearrangement • Ligand free • Regioselective • C-Benzoylation

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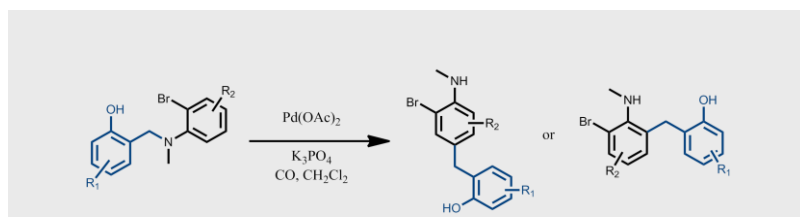
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**Palladium-Catalyzed Regioselective
C-Benzylation via a Rearrangement
Reaction: Access to Benzyl-
Substituted Anilines**

Rearrangement Reaction: A new C-Benzylation reaction was developed using $\text{Pd}(\text{OAc})_2$ and K_3PO_4 under a CO atmosphere. This transformation occurs in fine yields and in excellent regioselectivity, and constitutes a new way to obtain *para* or *ortho* benzyl-substituted N-methylanilines.

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